Lecture 1: Introduction to Evolutionary Algorithms

Introduction to Evolutionary Computation and Population-Based Search

Course: Evolutionary Algorithms

Prepared for classroom teaching

Instructor: Engr. Muhammad Farooq

Lecture roadmap and learning objectives

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Lecture flow

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1. Motivation and context

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2. What is optimization?

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- interpret a simple simulation

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- huge search spaces
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- many good trade-offs instead of one exact answer

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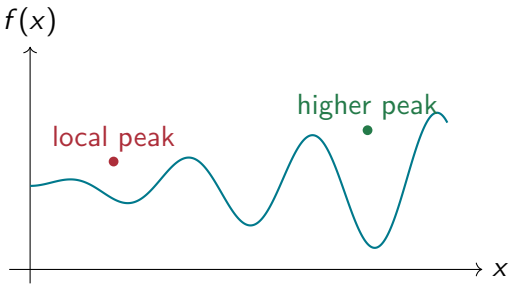
Typical goals

Minimize cost, distance, or time **or** Maximize profit, accuracy, or performance

A simple optimization example

Consider the function

$$f(x) = x \sin(10\pi x) + 1, \qquad x \in [-1, 2]$$



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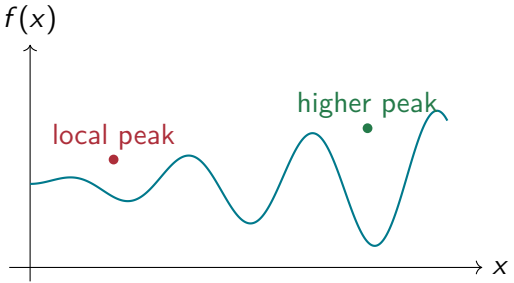

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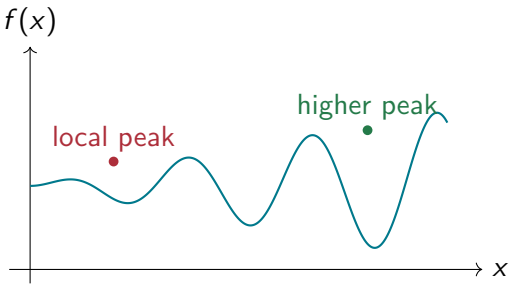

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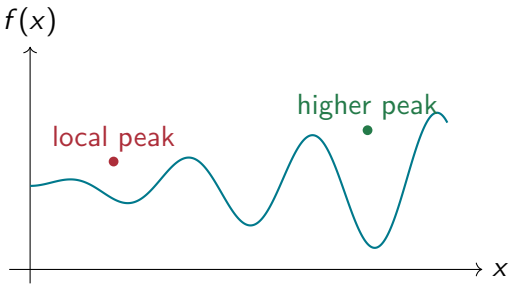

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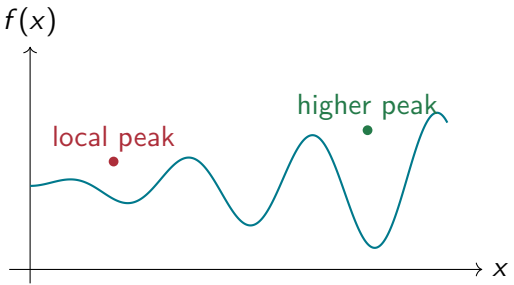

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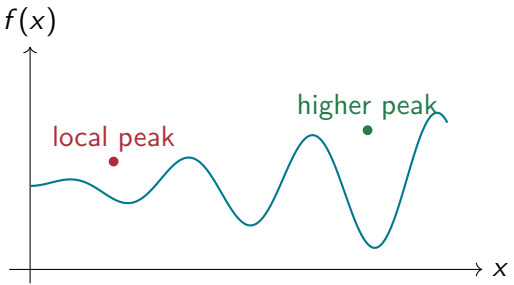

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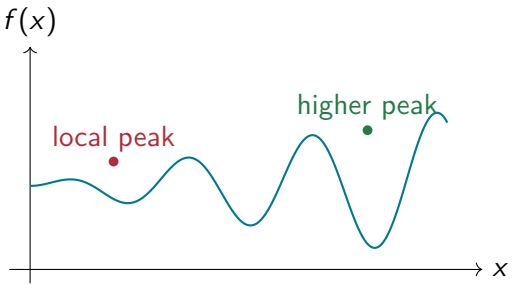

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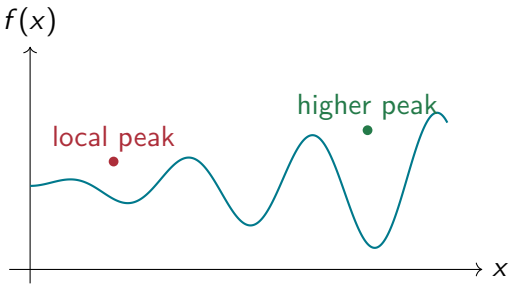

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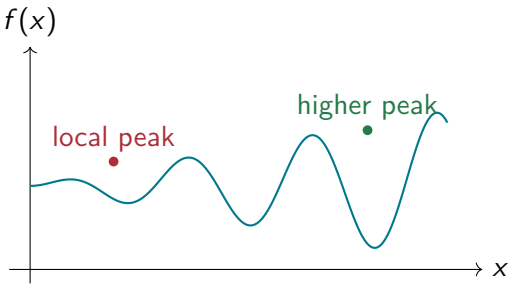

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Exhaustive search

Check every possible solution.

Problem: impossible when the search space is huge.

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possible binary strings of length 100.

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Hill climbing

Move from one solution to a better neighbor.

Problem: can get stuck in local optima.

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Local optimum vs global optimum

Transition point: Instead of exploring with one point only, evolutionary methods maintain *many* candidate solutions at once.

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Definitions

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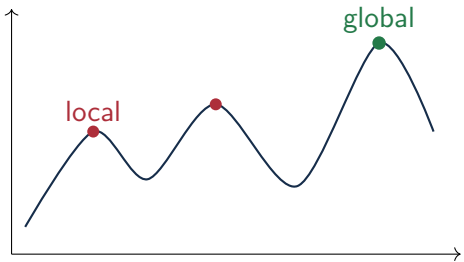

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Important note We do *not* simulate biology exactly; we borrow useful search principles from observed adaptive processes.

Biological inspiration vs algorithmic abstraction

Biology	Algorithm
organism	candidate solution
population	set of candidate solutions
environment	optimization problem
fitness	quality of a solution
reproduction	creating new solutions
mutation	random change
natural selection	choosing better solutions

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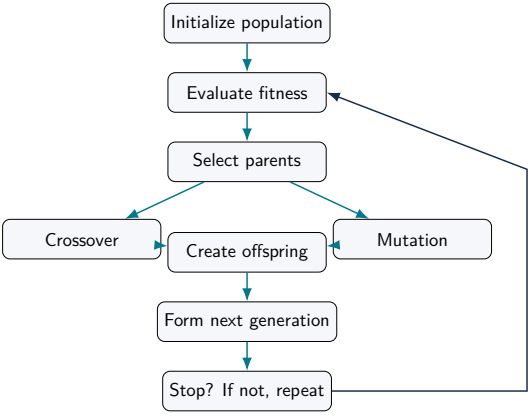
Common stopping criteria

Fixed number of generations, target fitness reached, or no improvement for many generations.

General workflow of an evolutionary algorithm



Important insight: Evolutionary algorithms are *stochastic*, *iterative*, and *approximate*—they do not guarantee the perfect solution every time, but they are often highly effective.

Random search vs hill climbing vs evolutionary search

Method	Main idea	Advantage	Limitation
Random search	Sample solutions randomly and keep the best	Very simple and easy to implement	Does not learn from previous good solutions
Hill climbing	Start with one solution and move to better neighbors	Often fast on easy landscapes	Easily trapped in local optima
Evolutionary algorithms	Maintain a population and improve it over generations	Better exploration and more robust search	More parameters and computational cost

Python simulation: objective function

```python
import numpy as np
import matplotlib.pyplot as plt

def f(x):
    return x * np.sin(10 * np.pi * x) + 1
```

Idea: We use a multimodal function with many peaks, so different search methods behave differently.

Python simulation: random search and hill climbing

```python
def random_search(n_samples=100):
    xs = np.random.uniform(-1, 2, n_samples)
    ys = f(xs)
    best_idx = np.argmax(ys)
    return xs, ys, xs[best_idx], ys[best_idx]

def hill_climb(start_x, step_size=0.05, max_iters=100):
    x = start_x
    history = [(x, f(x))]

    for _ in range(max_iters):
        candidates = [c for c in [x-step_size, x+step_size]
                      if -1 <= c <= 2]
        best_candidate = max(candidates, key=f)

        if f(best_candidate) > f(x):
            x = best_candidate
            history.append((x, f(x)))
        else:
            break

    return history
```

Python simulation: population-based sampling

```python
def population_sampling(pop_size=20, generations=20):
    best_history, avg_history = [], []
    population = np.random.uniform(-1, 2, pop_size)

    for _ in range(generations):
        fitness = f(population)
        best_history.append(np.max(fitness))
        avg_history.append(np.mean(fitness))

        idx = np.argsort(fitness)[::-1]
        survivors = population[idx[:pop_size // 2]]

        offspring = []
        while len(offspring) < pop_size:
            parent = np.random.choice(survivors)
            child = parent + np.random.normal(0, 0.1)
            child = np.clip(child, -1, 2)
            offspring.append(child)

        population = np.array(offspring)

    return best_history, avg_history, population
```

Python simulation: running the methods

```
np.random.seed(42)

rs_xs, rs_ys, rs_best_x, rs_best_y = random_search(n_samples=100)

start_x = np.random.uniform(-1, 2)
hc_history = hill_climb(start_x=start_x, step_size=0.05, max_iters=100)

best_hist, avg_hist, final_pop = population_sampling(
    pop_size=20, generations=20
)
print("Random Search Final x =", rs_best_x, "Final fitness =", rs_best_y)
print("Hill Climbing Final x =", hc_history[-1][0], "Final fitness =", hc_history
    [-1][1])
print("Population Sampling Final x =", final_pop[np.argmax(f(final_pop))],
      "Final fitness =", np.max(f(final_pop)))
```

Short conceptual examples for class discussion

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The number grows extremely quickly, so heuristic search becomes useful.

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Engineering design

If 10 parameters each have 100 possible values, then

$$100^{10}$$

possible designs must be considered.

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- run the program multiple times and compare outcomes

Next lecture: chromosome representation, selection, crossover, mutation, and the full genetic algorithm cycle.